

Sri Datta Ganesh Bandreddi

Machine Learning Engineer | deep learning, model training & optimization, and GPU acceleration

cosmobeans@cmu.edu • (412) 284-8386 • [linkedin.com/in/sri-datta-bandreddi](https://www.linkedin.com/in/sri-datta-bandreddi) • github.com/CosmoBean • cosmobeans.dev

EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00

Nagpur, India

Jul 2022

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash /Shell Scripting

DL Frameworks: PyTorch, JAX, TensorFlow, HF, NeMo/Megatron-LM, LoRA/QLoRA, FP8/BF16 Mixed Precision

GPU (Acceleration): CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

Data & ETL: Rapids (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML

Systems & MLOps: Docker, Kubernetes, Slurm, Jenkins/GitHub Actions, MLflow, Git, AWS (ECS, SageMaker), GCP

Data Infrastructure: PostgreSQL, Neo4j, Kafka, Redis, Iceberg, Elasticsearch, FAISS, MongoDB

PROFESSIONAL EXPERIENCE

MOTIONAL

Data Platform Intern (Infrastructure)

Pittsburgh, PA

May 2026 - Aug 2026

- Earned a provisional patent and accelerated scene-metadata retrieval, by building scene embeddings and an H3-backed similarity-search index over petabyte-scale drivetlogs
- Improved training-data quality and curated a retraining flywheel, by giving annotators a metadata platform backed by alternate source-of-truth signals and selecting long-tail events for retraining
- Cut drivetlog thaw costs by about \$100K/month, by building an agentic retrieval system that answers characterization and data-selection queries from metadata without thawing from cold storage

PHENOM

Product Development Engineer (Enterprise Talent Graph)

Hyderabad, India

Dec 2023 - Jul 2025

- Hardened a multi-tenant AI analytics platform for reliability under concurrent load, by leading six engineers to refactor shared services and reduce cross-tenant coupling
- Scaled a production knowledge graph past 1M entities with continuous ingestion, by designing validated aggregation pipelines on Neo4j schemas and Iceberg versioned storage
- Cut AI-driven query response time by 40% across 100+ clients, by tuning PostgreSQL sharding and hardening the Kafka platform (custom exporter, Kafka UI, ZooKeeper-to-KRaft migration)

BYJU'S

Software Engineer (Core Platform Team)

Hyderabad, India

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% for 1M+ users, by deploying scheduled AWS ECS auto-scaling and Terraform IaC tuned to traffic patterns
- Halved reliance on external identity providers, by leading development of an in-house auth platform (SAML, SSO, RBAC) in Go and Java (Spring Boot)

PHENOM

Machine Learning Intern

Hyderabad, India

May 2021 - Jun 2022

- Boosted recommendation accuracy for production pipelines, by building persona-classification models with spaCy and BERT embeddings across 3 data sources
- Cut recommender preprocessing latency, by migrating CPU pipelines to NVTabular on GPU across 2 production databases

PROJECT EXPERIENCE

FLARE: Efficient Attention (Carnegie Mellon University) (link)

Aug 2025 - Present

- Co-authored a linear-attention operator (3rd of 7 authors, submitted to TMLR) later adopted by an NVIDIA and General Motors crash-dynamics effort for roughly 2x lower memory
- Demonstrated over 200x speedups on Long Range Arena and extended FLARE to image classification and diffusion, by profiling multi-GPU NCCL training across FP8/BF16 with Nsight Systems

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B-parameter LLM to stable convergence with about 40% less memory on four H100 GPUs, by configuring tensor and pipeline parallelism, gradient checkpointing, and LoRA/QLoRA adapters
- Raised training throughput to 2.1x over naive data parallelism, by tuning NCCL, gradient bucketing, and optimizer-state sharding

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Reached 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification (beating ResNet-50, EfficientNet-B3, and ViT baselines), by training a 210M-parameter Swin Transformer with fused attention kernels
- Cut inference latency by 3x versus PyTorch eager, by converting to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and serving on Triton with adaptive batching

AWARDS & HONORS

Phenomenal Award for Outstanding Performance, Phenom (2024, Q1 & Q2) • 2nd Place, AppLovin Track, TartanHacks 2026 • Provisional Patent, Motional (2026)